Stronger spin-one gap bounds from the same finite certificates

Computational research note

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The exact Julia scalar calculations and the temperature-lifting argument below, combined with the analytic transfer and bootstrap statements in *The periodic spin-one Haldane gap*, give

$$\boxed{\gamma_L > 0.072901771 \quad \text{for every even } L \geq 36.}$$

Here $H_L = \sum_{j=1}^{L} \mathbf{S}_j \cdot \mathbf{S}_{j+1}$, $\mathbf{S}^2 = 2I$, and the bond coupling is one. The ground state is unique in this range. A simpler temperature-lifting choice gives the readable statement $\gamma_L > 7/100 = 0.07$ in the same range. The calculations reuse the existing trial vector, thermal contractions, shift, and polynomial filters. The new bound combines the stronger thermal data with the length-120 trial and raises inverse temperature before reducing the information to two purities.

Calculation	Gap lower bound	Even lengths covered
Published source 1	$\log(20)/784 \simeq 0.003821087$	$L \geq 2304$
Published source 2	$(4/105)\log(80/79) \simeq 0.000479192$	$L \geq 60$
Improved source 1	0.016978269	$L \geq 36$
Improved source 2	0.011977901	$L \geq 60$
Temperature lifting, simple	$0.070450002 > 7/100$	$L \geq 36$
Temperature lifting, optimized	0.072901771	$L \geq 36$

Source 1 uses the eighteen thermal values at lengths 4–12 and the length-72 trial certificate. Source 2 uses only nine thermal values at lengths 6, 8, and 10 and the length-120 trial certificate. Thus source 2 remains useful as the less expensive input set, even though the improved source-1 conclusion is stronger. The temperature-lifting calculation uses the eighteen source-1 thermal values and the length-120 trial; the original length-72 seed covers the early lengths. All these finite inputs were already reproduced. No global optimum is claimed.

1 Comparison with the numerical bulk gap

For the same convention $J = 1$, the large-system numerical estimates are

$$\Delta_{S=1} \simeq 0.41048(6), \qquad \Delta_{S=2} \simeq 0.08917(4).$$

These are the estimates in Table I of S. Todo and K. Kato, *Cluster Algorithms for General-S Quantum Spin Systems*, Phys. Rev. Lett. **87**, 047203 (2001). That table reports spatial correlation lengths 6.0153(3) and 49.49(1), respectively. The quoted numerical uncertainties are not rigorous spectral enclosures. The spin-one DMRG estimate 0.41050(2) of White and Huse, Phys. Rev. B **48**, 3844 (1993), is consistent with this scale.

The temperature-lifting lower bound is approximately 17.76% of the numerical bulk gap, compared with 4.14% for our earlier 0.016978269 bound and 0.93% for the published log(20)/784 bound. It improves our earlier bound by about 4.29 and the published bound by about 19.1, while remaining about 5.63 times below the numerical value. These statements compare a uniform bound on finite periodic chains with the numerical bulk benchmark; we have not independently reproduced the quoted thermodynamic-limit gap.

The selected quantity $A_j = (1-p_j)^2(1-q_j)$ below is a dimensionless *purity lower bound*, not a gap estimate. Along the successful bootstrap, where both length and inverse temperature grow as 2^j, the controlled defects tend to zero and $A_j \to 1$. The gap information is in the rate, through $\log[A_j/(1-A_j)]/b_j$, rather than the limiting value 1 itself. This is not a statement that purity tends to one at fixed temperature as length grows.

2 Input scope and provenance

The immutable upstream source is the OpenAI math repository, commit `adc7f1241b42e322a6451854ab7e4b4c146` directory `The-periodic-spin-one-Haldane-gap-September-24-2026` under `preprints`. The relevant arguments are in `build/initialization.tex` and `build/bootstrap.tex`. The previously completed Julia reproduction checks all finite thermal and trial integers against the upstream evidence. The new scalar calculation checks that reproduction's implementation and component hashes before using the input data. It does not describe reusing those outputs as fresh thermal contraction.

Let $a = 700741/500000$ and

$$Z_n(b) = \operatorname{Tr} e^{-b(H_n+naI)}, \qquad S(n,b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \qquad T(n,b) = \frac{Z_n(2b)}{Z_n(b)^2}.$$

The paper's transfer theorem supplies the real spatial eigenvalue lists and their sector multiplicities. Its length-72 and length-120 trial inequalities give $Z_{72}(b) > 1$ and $Z_{120}(b) > 1$ for every $b > 0$.

3 A compact improved source-1 initialization

Retain the original filters and their original error allowances:

$$\begin{aligned} F_J(x) &= x^4(7 + 17x - 280x^2 - 185x^3 + 1000x^4)^2, \\ F_N(x) &= x^4(12 - 394x^2 + 1000x^4)^2, \\ B_J &= 318604.54848175, \qquad B_N = 48169.880699. \end{aligned}$$

Every terminating decimal in a defining formula here denotes an exact rational. Choose

$$U = 1.001351, \quad V = 0.871256, \quad \ell = 0.999997, \qquad m_J = 1.0657205, \quad m_N = 0.2783155.$$

The verifier checks $F_J(\pm U) > B_J$ and $F_N(\pm V) > B_N$. It expands each $F(\pm(v+y))$ and checks every nonconstant coefficient is nonnegative. This verifies both entire exterior rays, so the paper's polynomial-filter lemma gives $|\lambda| < U$ in J and $|\lambda| < V$ in N.

Write

$$\mathcal{C}(m,h,r) = \lfloor m/h \rfloor h^r + (m - \lfloor m/h \rfloor h)^r, \qquad B(k) = 3\mathcal{C}(m_N, V^{12}, k/12).$$

The same capped-mass argument as in the paper now verifies

$$\mathcal{C}(m_J, \ell^{12}, 6) + B(72) < 1, \qquad \ell^{12} < m_J < 2\ell^{12}.$$

Since $Z_{72}(49/4) > 1$, one spatial eigenvalue in J has modulus greater than ℓ. Removing it leaves twelfth-moment mass at most $m_J - \ell^{12}$. Hence, for $k = 36, 72$,

$$R_A(k) = B(k) + (m_J - \ell^{12})^{k/12}$$

bounds the residual kth moment. At $(n, b) = (36, 49/4)$ we therefore have

$$1 - S(36, b) \le p_0 = 1 - (1 + R_A(36)/\ell^{36})^{-2},$$
$$1 - T(72, b) \le q_0 = 1 - (U^{72} + R_A(72))^{-2}.$$

Their numerical values are about 0.0449638076 and 0.176903757. These initial bounds need not satisfy the paper's common-defect bootstrap hypotheses. The coupled update permits unequal defects.

4 Cover the early lengths before applying the uniform theorem

The following finite-prefix version is a direct consequence of the paper's coupled update, purity interpolation, and uniform bootstrap proposition.

Proposition 1. *Suppose p_j, q_j are valid upper bounds on the spatial and temporal defects at $(n_j, b_j) = (2^j n_0, 2^j b_0)$ for $0 \le j \le K$. Put $A_j = (1 - p_j)^2(1 - q_j)$ and suppose $A_j > 1/2$ for $j < K$. Suppose $u = \max(p_K, q_K)$ and C satisfy*

$$0 < u < 1/6, \quad G(u) := \tfrac{1}{2}\big((1-u)^{-1} + (1-u)^{-2} + (1-u)^{-3}\big)^2 \le C, \quad Cu < 1, \quad C(1-3u) > 3.$$

If g is strictly smaller than every member of

$$\left\{ \frac{\log(A_j/(1-A_j))}{b_j} : 0 \le j < K \right\} \cup \left\{ \frac{\log(1/(Cu))}{b_K} \right\},$$

then the ground state is unique and $\gamma_L > g$ for every even $L \ge n_0$.

Proof. For $n_j \le L \le 2n_j$, the interpolation inequality gives $T(L, b_j) \ge A_j^{L/(2n_j)} \ge A_j > 1/2$. The largest physical Gibbs weight w_0 is at least this purity, so it is unique. For a weight w_1 at the first excited energy,

$$e^{-b_j \gamma_L} = w_1/w_0 \le (1 - w_0)/w_0 \le (1 - A_j)/A_j.$$

This proves the required estimate on each finite interval. The original uniform bootstrap proposition covers every even $L \ge n_K$. The intervals $[n_j, 2n_j]$ for $j < K$ and the last half-line cover all even lengths at least n_0. □

Use the original update $f(u) = u^2/[2(1-u)^2]$, retaining the two defects:

$$p_{j+1} = \big\lceil f(1 - (1 - p_j)^2(1 - q_j)) \big\rceil_{100},$$
$$q_{j+1} = \big\lceil f(1 - (1 - q_j)^2(1 - p_{j+1})) \big\rceil_{100}, \qquad \lceil x \rceil_{100} = 10^{-100} \lceil 10^{100} x \rceil.$$

For the compact constants, take $K = 9$ and $C = 5$. The verifier checks all uniform-theorem hypotheses and all intervening length intervals. The following entries are explanatory approximations; full rational bounds are in the certificate.

j	p_j	q_j	$\log(A_j/(1-A_j))/b_j$
0	0.0449638076	0.176903757	0.0900056767
1	0.0551175965	0.158002808	0.0452204114
2	0.0545331044	0.120970423	0.0265228864
3	0.0371649174	0.0592125617	0.0195936555
4	0.0107429704	0.0100975890	0.0175199506
5	0.0005203714	0.0002212682	0.0170259785
6	$7.97715\,10^{-7}$	$9.83379\,10^{-8}$	0.0169496858
7	$1.43443\,10^{-12}$	$1.93410\,10^{-14}$	0.0169454007
8	$4.17084\,10^{-24}$	$7.48147\,10^{-28}$	0.0169453721

At $j = 9$, $n_9 = 18432$, $b_9 = 6272$, and $u \simeq 3.47981322\,10^{-47}$. The uniform tail bound exceeds $0.016799278 > 1/60$. All earlier interval bounds exceed $1/60$ as well. This proves the compact claim.

5 Tighter constants and the less expensive branch

Exact bisection on a grid of spacing 10^{-12} gives valid source-1 choices

$$U = 1.001350644012, \quad V = 0.871255704653, \quad \ell = 0.999997937629.$$

Eleven coupled updates, upward rounding at spacing 10^{-512}, and an outward-rounded $C \geq G(u)$ then establish the earlier improved bound 0.016978269. The very fine rounding avoids a floor in the rapidly squaring defects; it concerns only scalar integers, not tensor dimensions.

For source 2 the script preserves the filters $x^6(-2+10x^2)^2$ and $x^6(-25+100x^2)^2$, the thermal error 10^{-5}, the moment degrees, and the temperature multiplier $5/2$. It sharpens the two cutoffs to $U = 1.002408155316$, $V = 0.886411420539$, uses the sector-center-plus-error moment bounds directly, and obtains rational upper bounds

$$D = 1.445289213010, \quad E = 1.198064585836, \quad R_q = 0.064761703869.$$

As in the original proof, $D^2 > m^5$, $E^2 > p^5$, and $R_q^2 > (q-1)^5$. Set $\tau = (D+11E)/12$ and $\tau_0 = (D+3E)/4$. The original four-vertex residual bound $R(t)$ is repeatedly used to upgrade the dominant weight whenever $s^4 + R(t)^2 < 1$. The final lower bound is $t = 0.998895181835$; the resulting initial defects are approximately 0.1209336963 and 0.1179460383. Applying the same finite-prefix proposition gives $\gamma_L > 0.011977901$ for every even $L \geq 60$.

For comparison, merely retaining the original exact seeds and improving the bootstrap gives 0.004710839 at $L \geq 36$ from source 1 and 0.006924382 at $L \geq 60$ from source 2. Thus both tighter initialization and better use of the coupled recurrence contribute to the improvement.

6 Temperature lifting before the purity bootstrap

The following recombination of the same finite inputs gives a larger gain. It generalizes the paper's source-2 temperature argument to the stronger source-1 moments, retaining the threefold sector multiplicity at every required moment order.

Use the tighter rational U, V, m_J, m_N above and put $b_0 = 49/4$. For every even $k \geq 12$, capped masses give

$$\begin{aligned} \mathcal{J}(k) &= U^k + (m_J - U^{12})^{k/12}, \\ \mathcal{N}(k) &= V^k + (m_N - V^{12})^{k/12}, \\ Z_k(b_0) \leq M_k &:= \mathcal{J}(k) + 3\mathcal{N}(k), \qquad Z_k(b_0, P) \leq \mathcal{J}(k). \end{aligned}$$

Indeed $U^{12} < m_J < 2U^{12}$ and $V^{12} < m_N < 2V^{12}$. The last inequality uses $Z_k(b_0, P) = J_k - N_k$ and $N_k \geq 0$.

Choose $r \geq 1$ and an even $12 \leq k \leq 60$. Because each twisted partition function is itself the Gibbs trace of a self-adjoint Hamiltonian, raising its positive Boltzmann factors to the power r gives

$$Z_k(rb_0) \leq D := M_k^r, \qquad Z_k(rb_0, P) \leq E := \mathcal{J}(k)^r.$$

Outward rational upper bounds may replace D, E. In the new spatial operator at $b = rb_0$, set $w_i = |\lambda_i(b)|^k$ and

$$h = 60/k \geq 1, \qquad \tau = (D + 3E)/4.$$

Then $\sum_i w_i \leq D$, the J-sector mass is at most τ, and the existing length-120 trial gives

$$\sum_i w_i^{2h} = Z_{120}(b) > 1.$$

Choose a rational $t > 0$ with $Dt^{2h-1} < 1$ and $3t > D$. Some weight $w_0 > t$ exists, since otherwise the last sum is at most $Dt^{2h-1} < 1$. This weight belongs to J: every N weight occurs three times, which would contradict the total mass bound. After removing w_0, let x be the remaining J mass. Then $0 \leq x \leq \tau - t$, while the full remaining N mass is at most $D - t - x$. For every $s \geq 1$, the residual sth moment is at most

$$R_s(t) = \max\left\{3\left(\frac{D-t}{3}\right)^s, (\tau - t)^s + 3\left(\frac{D-\tau}{3}\right)^s\right\}.$$

To prove this, bound each sector list by the sth power of its total mass. The three identical N lists contribute at most $3[(D - t - x)/3]^s$. The resulting convex function of x attains its maximum at one of the two endpoints. All quantities are nonnegative because $t \leq \tau \leq D$.

We may repeatedly replace t by a larger rational t' whenever

$$(t')^{2h} + R_{2h}(t) < 1.$$

If $w_0 \leq t'$, this inequality would contradict $Z_{120}(b) > 1$. Using R_{2h} directly preserves more multiplicity information than squaring an upper bound on R_h. Consequently a spatial seed at length 60 is

$$S(60, b) \geq \left(1 + \frac{R_h(t)}{t^h}\right)^{-2}.$$

The temporal seed uses the same trial in the physical spectrum. Write $Q = M_{120} > 1$. At b_0, the ground Boltzmann factor $z_0 > 1$ and the total mass is at most Q. Hence

$$\sum_{i>0} \frac{z_i}{z_0} \leq Q - 1, \qquad \sum_{i>0} \left(\frac{z_i}{z_0}\right)^r \leq (Q-1)^r.$$

This yields the second seed without a new contraction:

$$T(120, b) \geq \left(1 + (Q-1)^r\right)^{-2}.$$

The spatial weights w_i and physical Boltzmann factors z_i play different roles; the proof uses the original transfer theorem to connect their partition functions.

k	r	$1 - S(60, rb_0)$ upper	$1 - T(120, rb_0)$ upper	Gap bound
30	2	0.0307608679	0.0590740709	0.070450002
28	17/8	0.0305302052	0.0479556776	0.072901771

The displayed defect bounds are rounded upward for readability; the calculation uses full rational enclosures. Ten coupled updates use an upward rounding grid 10^{-2048}, keeping its floor below the relevant defects, followed by the uniform tail theorem. This covers even $L \geq 60$. The earlier source-1 seed gives an interval bound greater than 0.09 on every even $36 \leq L \leq 72$. Combining these overlapping ranges establishes both displayed bounds for every even $L \geq 36$. The simpler row just doubles inverse temperature and uses $h = 2$; its residual estimates therefore involve only second and fourth powers. The numerical discovery script scans even k from 12 to 60 and r from 1 to 3 in increments of 1/40. Its output is not a certificate; the two selected parameter choices are replayed through the rational verifier.

7 Reproduction and the arithmetic trust boundary

Run from the repository root:

```
julia --startup-file=no --project=spin1 \
  spin1/experiments/optimize_bounds.jl
julia --startup-file=no --project=spin1 \
  spin1/experiments/test_bounds.jl
julia --startup-file=no --project=spin1 \
  spin1/experiments/temperature_lift.jl
julia --startup-file=no --project=spin1 \
  spin1/experiments/test_temperature_lift.jl
```

The output `spin1/experiments/bounds.json` contains the complete initialization parameters, every rational defect pair, interval coverage, the tail-theorem parameters, logarithm enclosures, and source hashes. All accepting comparisons use integers or rationals. Displayed floating-point numbers are diagnostic only. The new `temperature-lift.json` records the lifted moment bounds, dominant-weight refinements, both bootstrap chains, and the early-length bridge, together with script and input hashes. Fractional powers are enclosed using integer roots: for $x \geq 0$ and $r = a/b \geq 0$, the verifier checks rational endpoints l, u through $l^b \leq x^a \leq u^b$. No floating-point comparison accepts a certificate.

To enclose a logarithm, range-reduce $x = 2^k y$ with $1 \leq y < 2$, round y outward to a rational grid, and use $z = (y-1)/(y+1)$ in

$$\log y = 2 \sum_{j=0}^{M-1} \frac{z^{2j+1}}{2j+1} + E_M, \qquad 0 \leq E_M \leq \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

Here $M = 40$ and $0 \le z \le 1/3$; $\log 2$ is treated by the same formula. The final rational claim is strictly below all computed lower endpoints.

The tests independently rearrange the coupled updates, verify continuous coverage of lengths, check the seed inequalities and reject unsafe cutoffs, rebuild all 27 input thermal enclosures, and check both variational signs. They also use high-precision floating-point logarithms as an independent numerical control, not as certificate authority. This is exact finite arithmetic combined with the paper's analytic arguments; it is not a Lean formalization of the transfer theorem or of the infinite induction. The physical trial still has bond dimension 26.

8 First exact spin-two variational inputs

The spin-two extension uses $\mathbf{S}^2 = 6I$ and the same coupling $J = 1$. The scalar coupled recurrence and the finite-prefix argument above apply once its model-specific thermal and variational inputs have been supplied. The smaller numerical gap and longer correlation length make those inputs more demanding; the spin-one gap constant cannot simply be rescaled.

The new Julia search uses virtual space

$$V_0 \oplus 2V_1 \oplus 2V_2 \oplus V_3, \qquad D = 24,$$

with 29 real reduced coefficients. Exact highest-weight equations construct integer polynomial tensors. After rounding the numerical coefficients to integers, the physical matrices have the form

$$A_m = \sqrt{\binom{4}{2+m}}\, P_m, \qquad P_m \in \mathbb{Z}^{24\times 24}, \quad -2 \le m \le 2.$$

The square roots cancel in the norm and bond-energy contractions, leaving integer transfer blocks. Their largest dimension is 106. The finite certificate verifies

$$\frac{\langle \psi_N, H_N \psi_N \rangle}{N \langle \psi_N, \psi_N \rangle} < -4.7595 \qquad (N = 128, 256, 512).$$

The corresponding energy densities, shown only as approximations, are -4.7595998798, -4.7595993033, and -4.7595993033. Thus $E_0(N) < -4.7595N$ and the shifted partition function exceeds one at each of these lengths for every positive inverse temperature. Direct four-site integer amplitudes agree with the transfer contraction; an independent ITensors MPO/MPS calculation agrees in total energy within 7.2×10^{-15}.

The spin-two numerical bulk energy density in Todo–Kato is $-4.761249(6)$, so this first trial is still roughly 0.00165 per bond above that reference. Its energy certificate does not certify a gap. No spin-two finite-temperature purity seed has yet been certified. The existing periodic DMRG pilot gives gaps near 0.54557 at $L = 8$ and 0.3109 at $L = 16$; the latter has appreciable residuals. These short-chain numbers should not be used as estimates of the 0.08917 bulk gap.

The scalar screening has also been relaxed from requiring both purities to exceed 0.99. Exact recurrence checks admit, for example, target purities $(S, T) \ge (0.95, 0.83)$. If such targets were realized, the necessary energy screen would be

$$\frac{\log[T/(1-T)]}{\Delta_{2n}} \le \beta \le \frac{-\log(ST)}{E_0(2n) - 2E_0(n)},$$

for positive denominator. This permits the energy-mismatch-to-gap ratio to be as large as 0.14986, versus 0.00437435 for the old 0.99 targets. It is only a necessary screen; neither target pair has been established for spin two, and the current small-chain pilot fails even the relaxed screen.

The scripts `spin2/scripts/certify_trial.jl` and `spin2/scripts/bootstrap_screen.jl` reproduce these two steps. The full integer trial parameters, norms, energy numerators, and block contributions are retained in `spin2/results/trial-certificate.json`.